The Systems Behind Modern AI

Why and how does hardware designed for 3D games power modern AI

Sam Ly

Cal Poly Pomona

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How does Matrix Multiplication map to Neural Networks?

Why GPUs?

Optimizing **Ge**neral Matrix Multiplication (GEMM)

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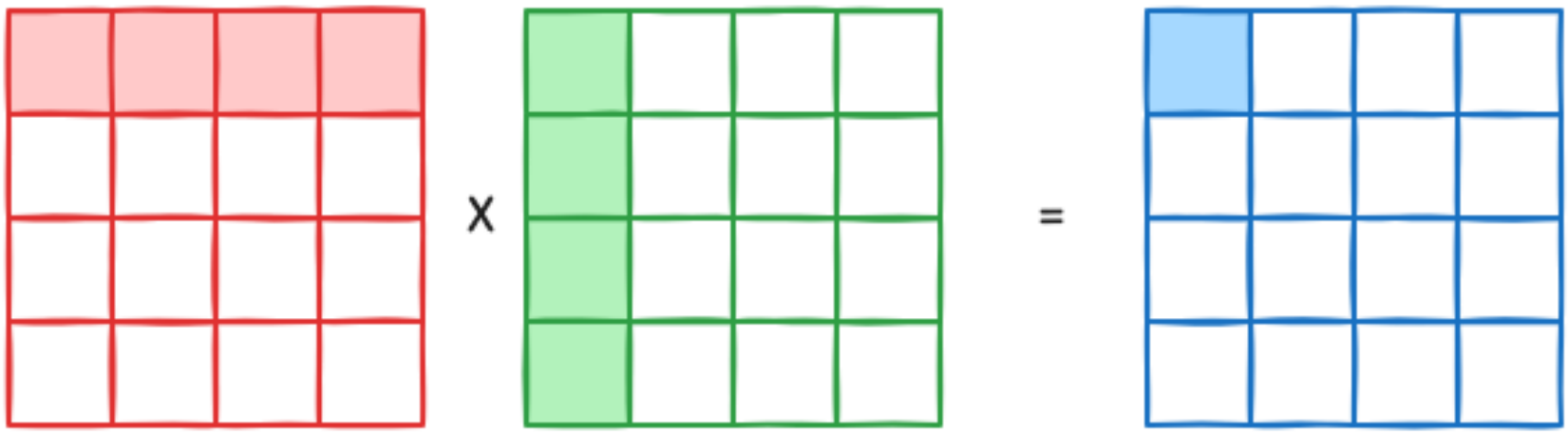
History of GPUs

- 1999: GeForce 256
- 2004: Brook Programming, more complex "rendering" pipelines, GP-GPU
- 2006: GeForce 8800, Compute Unified Device Architecture (CUDA)
- 2008: Fermi, General Purpose Scientific Compute becomes first class, L1/L2 caches
- 2010-2016: incremental improvements to power efficiency, programming models, and multi-GPU systems.
- 2017: Volta, AI-specialization: tensor cores designed around deep learning primitives
- 2022: Hopper, Transformer Engine, further specializing into AI
- 2026: Since 2022, we have only really seen incremental improvements.

Parallelism

Graphics rendering is a naturally parallel task. Each pixel does the same work.
CPUs can't keep up, so we trade logical capability for **throughput**.
Each *thread* on the GPU can only do simple things, but we have many of them. GPU's can't do some of the more algorithmically complex tasks, but can do a a large amount of simple tasks much faster.
That is, a CPU has few threads that are very capable. A GPU has many dumb threads.

Matrix Multiplication is Parallel



The operations are very simple, but the number of operations grow on the order of $\mathcal{O}(n^3)$.

CPU implementation

```
void matmul(const std::vector<double>& a,
            const std::vector<double>& b,
            std::vector<double>& c,
            std::size_t n) {
    for (std::size_t i = 0; i < n; ++i) {
        for (std::size_t j = 0; j < n; ++j) {
            double sum = 0.0;
            for (std::size_t k = 0; k < n; ++k) {
                sum += a[i * n + k] * b[k * n + j];
            }
            c[i * n + j] = sum;
        }
    }
}
```

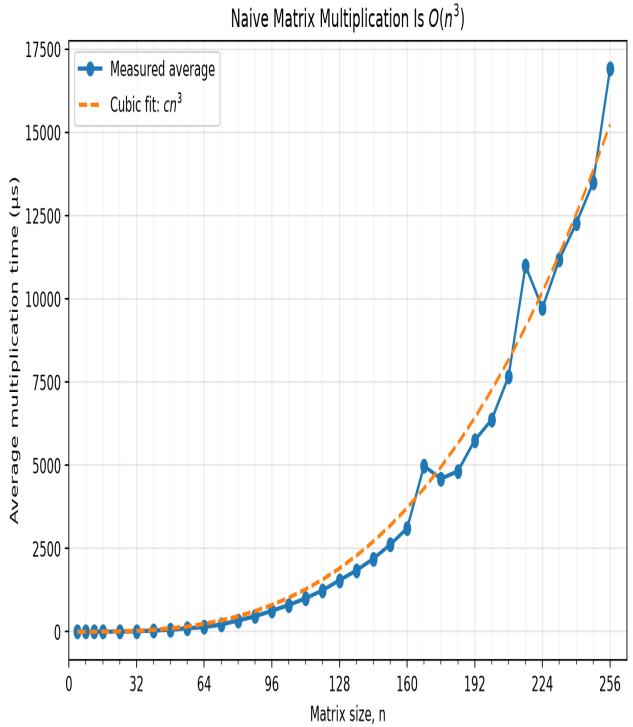


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Naive

Arithmetic Intensity, throughput, and the Roofline Model